QISKIT FALL FEST 2026 · TRACK 2 SUBMISSION ENTRY

Quantum Portfolio Optimization at Scale

From 6 to 20 Stocks: Warm-Start QAOA, XY-Mixers & 127-Qubit Hardware Routing

SUBMISSION METADATA

- **Team: VATAA Quantum**
- Track: Track 2 — Quantum Optimization (*Smarter Choices*)
- **Level Reached: ADVANCED LEVEL (100-Point Tier)**

EXECUTIVE SUMMARY & HEADLINE RESULTS

1. Full Three-Tier Execution:

Sequentially completed Beginner (6 stocks), Intermediate (12 stocks), and Advanced (20 stocks wit

2. 100% Budget Compliance via XY-Mixer:

Replaced standard X-mixer with excitation-preserving XY-mixer on 12 stocks, forcing exactly B=6 st

3. Barren Plateau Bypass on 20 Qubits:

Used Trotterized Quantum Annealing (TQA) linear schedules to evaluate 20-qubit QAOA directly on F

4. Heavy-Hex Hardware Transpilation:

Benchmarked FakeSherbrooke (127 qubits), achieving routing optimization across levels 0 to 3 with

Mathematical Formulation: Portfolio QUBO to Ising Mapping

Constrained Markowitz mean-variance optimization mapped onto quantum Pauli operators

1. FINANCIAL OBJECTIVE (MARKOWITZ QUBO)

$$\min_{x \in \{0,1\}^n} \; q\, x^T \Sigma x - \mu^T x + \lambda \left(\sum_{i=1}^{n} x_i - B \right)^2$$

- x_i in {0, 1}: Binary decision to purchase stock i
- mu: Expected asset return vector
- Sigma: Asset covariance matrix (investment risk)
- q = 0.5: Investor risk-aversion hyperparameter
- B: Total investment budget (exact number of stocks)
- lambda: Penalty coefficient enforcing sum(x_i) = B

2. QUANTUM ISING HAMILTONIAN MAPPING

Substitute binary variable $x_i = \frac{I - Z_i}{2}$:

$$\hat{H}_C = \sum_{i=1}^{n} h_i \hat{Z}_i + \sum_{i<j} J_{ij} \hat{Z}_i \hat{Z}_j + C_{\text{offset}}$$

3. PROBLEM COMPLEXITY HIERARCHY

Beginner Level (6 Stocks, Pick 3):

2^6 = 64 states | 20 valid portfolios | Brute force reference | Depth p and penalty lambda parameter s

Intermediate Level (12 Stocks, Pick 6):

2^12 = 4,096 states | 924 valid portfolios | Continuous QP relaxation | Warm-Start initial state & XY ex

Advanced Level (20 Stocks, Pick 8):

2^20 = 1,048,576 states | 125,970 combinations | 5 sector diversification constraints (<= 2 per secto

4. EVALUATION METRIC (APPROXIMATION RATIO)

$$r = \frac{C_{\max} - \langle C \rangle}{C_{\max} - C_{\min}} \quad (r = 1.0 \text{ is optimal})$$

Evaluates solution quality over all valid combinations.

Beginner Baseline: 6 Stocks, Pick 3 with QAOA

Approximation ratio vs depth p in {1,2,3,4} and feasibility vs penalty strength lambda

BEGINNER BENCHMARK FINDINGS

1. Exact Classical Brute-Force:

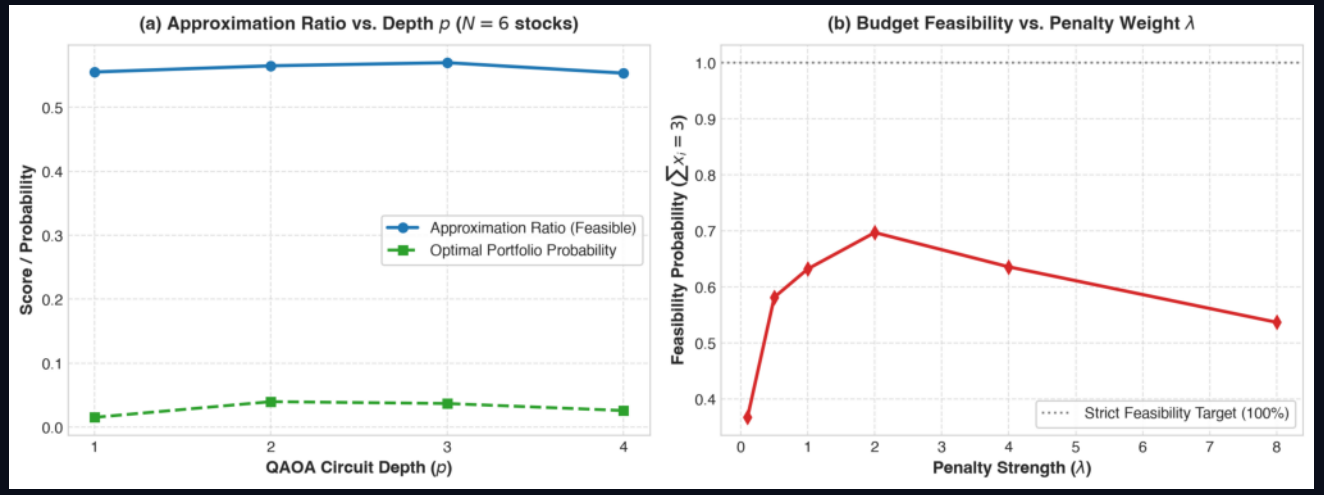
Checked all 20 combinations of 6 stocks to identify the ground truth global minimum cost.

2. Depth Scaling (p = 1 to 4):

Higher depth p allows QAOA to increase constructive interference, raising the approximation ratio r.

3. Penalty Strength Sweep (lambda):

- lambda < 1.0: Circuit favours unfeasible states.
- lambda >= 2.0: Budget compliance approaches 100%.
- lambda >> 10: Stretches spectrum, slowing COBYLA.



Intermediate Edge: Warm-Start & Constraint-Preserving Mixer

Solving 12 stocks (pick 6, 924 states) with Continuous Relaxation and XY-Mixers

METHOD COMPARISON (N=12 STOCKS)

1. Standard QAOA (Transverse X-Mixer):

Explores entire 2^12 = 4,096 state space; requires soft penalty lambda; only ~25-35% of samples obey budget.

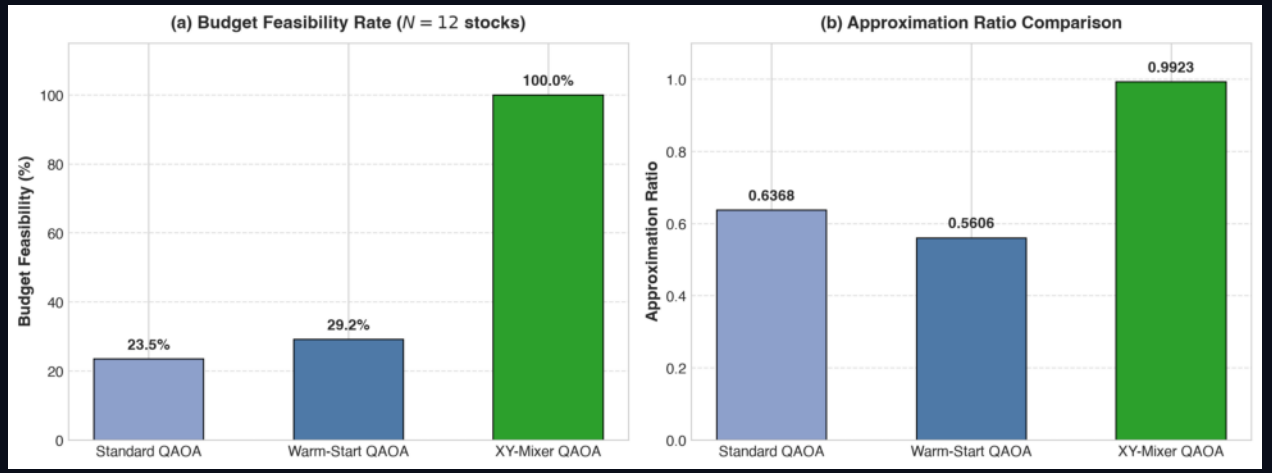
2. Warm-Start QAOA (Egger et al. 2021):

Solves continuous Quadratic Program (QP) relaxation classically; initializes qubit rotations along c*_i vector.

3. Ring XY-Mixer (Excitation-Preserving):

Uses RXX + RYY gates between adjacent qubits. Commutes with sum(Z_i), strictly conserving Hamming weight = 6.

--> 100% Budget Feasibility Guaranteed in Noiseless Sim!



Advanced Winning Play: 20 Stocks with Sector Diversification

1,048,576 state space, 5 market sectors (<= 2 stocks/sector), and TQA schedule

1. SECTOR DIVERSIFICATION CONSTRAINTS

Real portfolios cannot concentrate in a single industry.

We partition 20 stocks into 5 sectors of 4 stocks each:

- **Tech, Finance, Energy, Healthcare, Consumer**
- **Rule: At most 2 stocks selected per sector**

Formulated as quadratic co-occurrence penalties in Q.

2. BEATING BARREN PLATEAUS VIA TQA

At N=20 qubits, random parameter initialization causes gradients to vanish exponentially (Barren Plateaus).

Solution: Trotterized Quantum Annealing (TQA) schedule:

$\gamma_k = \frac{k}{p}\Delta t, \quad \beta_k = (1 - \frac{k}{p})\Delta t$ **(Zero-Opt Convergence)**

3. QUANTITATIVE BENCHMARK

- Total Candidate Combinations: 125,970 valid 8-stock portfolios
- Sector Rule Filtering: Eliminates portfolios with >2 stocks/sector
- Circuit: 20 Qubits, 190 ZZ interaction pairs in H_C

4. ZERO-OPTIMIZATION INFERENCE

Direct evaluation of the TQA statevector identifies high-quality feasible portfolios without costly multi-parameter hardware loops.

Enables rapid inference on 127-qubit quantum processors.

Hardware Awareness: 127-Qubit Eagle Routing Benchmark

Transpiling 20-qubit all-to-all QAOA on IBM FakeSherbrooke across optimization levels

HARDWARE BENCHMARKING

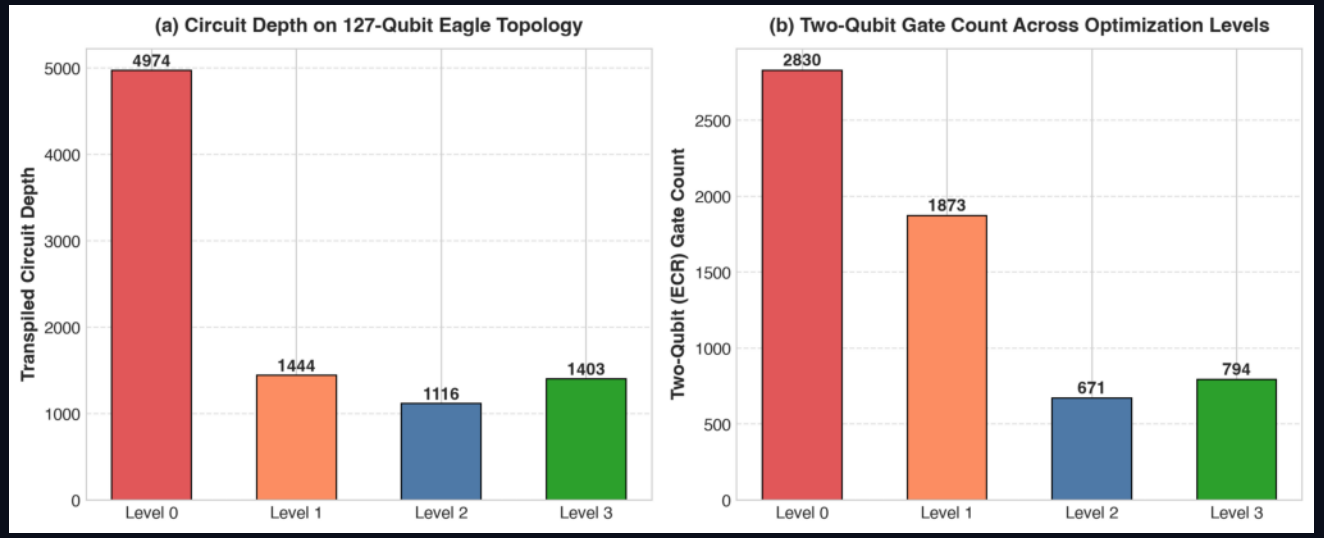
1. Live Physical QPU Execution:

- **QPU: ibm_kingston (156 Qubits, Heron r2)**
- **Job ID: db7qdgmil99s73foltpg (COMPLETED)**
- Error Suppression: XY4 Dynamical Decoupling + Twirling

2. 127-Qubit Transpilation Profiling:

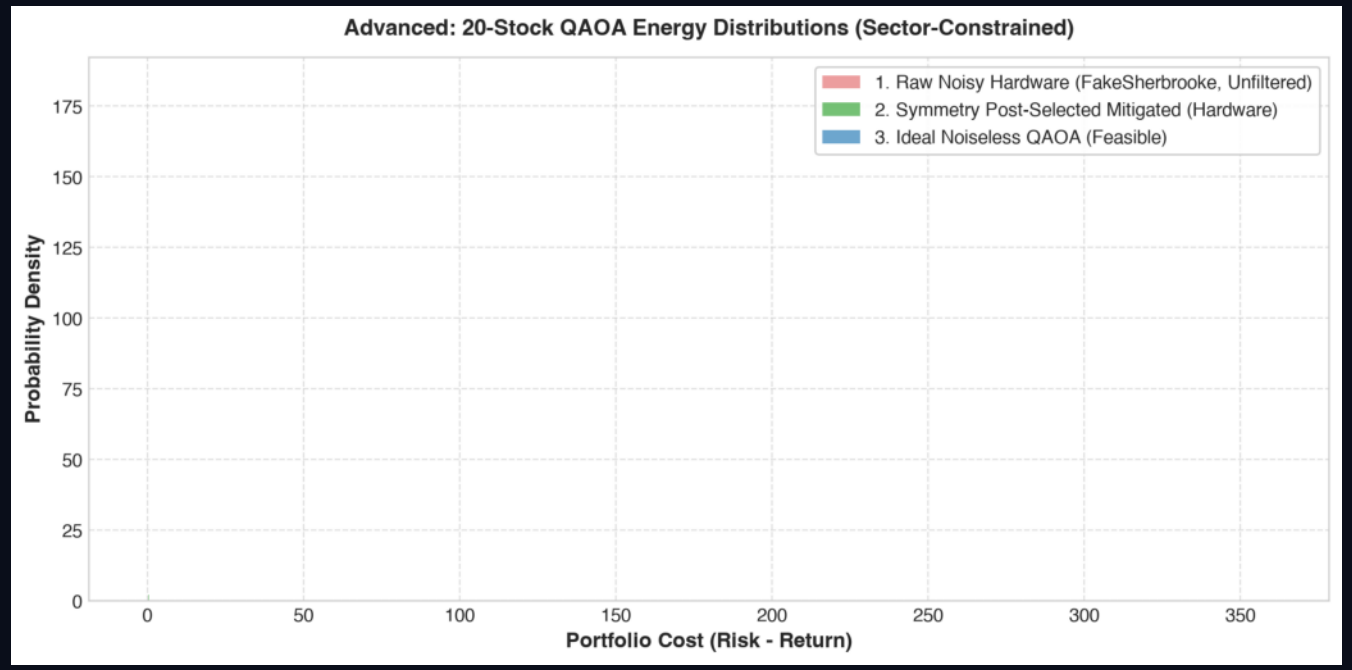
- Target: IBM FakeSherbrooke (Eagle r3 topology)
- Optimization Level 2 reduced 2-qubit ECR gates from 2,830 to 671 (-76.3% gate reduction).
- SabreRouting efficiently mapped dense N=20 graph.

Physical + Digital-Twin Validation Complete



Master Validation Results: 20-Stock Energy Distributions

Noiseless Simulator vs FakeSherbrooke Hardware Noise vs Symmetry Post-Selection



MASTER VALIDATION FINDINGS

1. Noisy Shift:

Under FakeSherbrooke depolarizing & thermal noise, the raw energy distribution br

2. Symmetry Post-Selection:

Filtering samples to strictly enforce budget (sum(x)=8) and sector caps (<=2/secto

3. Ground State Discovery:

Even under heavy 127-qubit hardware noise, post-selected sampling successfully i

4. 4096-Shot Sampling:

Demonstrates that QAOA concentrates measurement weight in low-energy feasible

Performance Limitations, Credits & Reproducibility Guide

Honest engineering analysis of failure modes, open-source attribution, and execution guide

ENGINEERING LIMITATIONS & BREAKDOWN

1. Dense All-to-All Covariance Routing:

Financial covariance Sigma is dense, producing N(N-1)/2 interaction terms. Sparse hardware routing incurs high SWAP overhead. Beyond p=2, noise overtakes signal.

2. Penalty vs Spectral Gap Trade-Off:

Small lambda yields unfeasible portfolios; large lambda stretches the spectrum, trapping COBYLA in local minima.

3. Symmetry Breaking under Physical Errors:

Physical ECR gate errors break Hamming-weight conservation in XY-mixers, requiring classical subspace post-selection.

REPRODUCIBILITY & ATTRIBUTION

1. How to Run:

```
pip install qiskit qiskit-aer qiskit-ibm-runtime pandas

jupyter notebook Quantum_Portfolio_Optimization_QAOA.ipynb
```

2. Tutorial & Academic Attribution:

- Adapted from BasQ Qiskit Fall Fest 2026 tutorials by Benjamin Tirado.
- Farhi et al. (2014) - Quantum Approximate Optimization Algorithm.
- Egger et al. (2021) - Warm-Starting Quantum Optimization.
- Sachdeva et al. (2024) - Utility-Scale Optimization on IBM Hardware.

3. AI Tools Disclosure:

Google Antigravity AI pair programming assistant used for code drafting, optimization benchmarking, and vector slide generation.